



ST BENEDICT'S

SUBJECT
GRADE
EXAMINER
NAME
TEACHER

Mathematics
11
Miss A Balzer

PAPER
DATE
MARKS
MODERATOR
DURATION

Paper 1
2 August 2019
130
Gr 11 Teachers
2 ½ hours

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Algebraic Expressions	16	
2	Algebraic Equations	36	
3	Nature of Roots	13	
4	Problem Solving	7	
5	Sequences and Series	21	
6	Basic Graphs	18	
7	Graph Interpretation	19	
TOTAL		130	

pg 1-2
pg 5-7
pg

QUESTION 1

16 MARKS

a) Simplify the following:

$$\begin{aligned}
 1) \quad & \sqrt{a^3} \cdot \left(a^{-\frac{1}{4}}\right)^2 \\
 &= a^{3/2} \cdot a^{-1/2} \\
 &= a \quad \checkmark
 \end{aligned}
 \tag{3}$$

$$\begin{aligned}
 2) \quad & \frac{12^{n+1} \cdot 27^{n-2}}{18^{2n-1}} \\
 &= \frac{(3 \cdot 2)^{n+1} \cdot (3^3)^{n-2}}{(3^2 \cdot 2)^{2n-1}} \\
 &= \frac{3^{n+1} \cdot 2^{2n+2} \cdot 3^{3n-6}}{3^{4n-2} \cdot 2^{2n-1}} \\
 &= 3^{n+1+3n-6-4n+2} \cdot 2^{2n+2-2n+1} \\
 &= 3^{-3} \cdot 2^3 \\
 &= \frac{8}{27} \quad \checkmark
 \end{aligned}
 \tag{5}$$

$$\begin{aligned}
 3) \quad & \frac{3^{x+1} - 3^x}{3^{x+2} - 2 \cdot 3^x} \\
 &= \frac{3^x \cdot 3 - 3^x}{3^x \cdot 3^2 - 2 \cdot 3^x} \\
 &= \frac{\cancel{3^x} (3 - 1)}{\cancel{3^x} (3^2 - 2)} \\
 &= \frac{2}{7} \quad \checkmark
 \end{aligned}
 \tag{4}$$

Trene

b) Rationalise the denominator:

1) $\frac{10}{3-\sqrt{2}}$

(4)

$$= \frac{10}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} \checkmark$$

$$= \frac{30 + 10\sqrt{2}}{9-2} \checkmark$$

$$= \frac{30 + 10\sqrt{2}}{7} \checkmark$$

QUESTION 2

36 MARKS

a) Solve for the variables in each of the following:

1) $6x^2 - 19x + 10 = 0$

(3)

$$(2x-5)(3x-2) = 0 \checkmark$$

$$x = 5/2 \checkmark \text{ or } x = 2/3 \checkmark$$

2) $3x^2 - 4x = 5$, Leave answer in simplest surd form.

(4)

$$3x^2 - 4x - 5 = 0 \checkmark$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-5)}}{2(3)} \checkmark$$

$$x = \frac{4 + \sqrt{76}}{6} \checkmark \text{ or } x = \frac{4 - \sqrt{76}}{6} \checkmark$$

$$\frac{2 + \sqrt{19}}{3}$$

$$\frac{2 - \sqrt{19}}{3}$$

Christo

3) $\sqrt{2-x} - 4 = x$

(6)

$$\sqrt{2-x} = x + 4 \checkmark$$

$$2-x = (x+4)^2 \checkmark$$

$$2-x = x^2 + 8x + 16$$

$$0 = x^2 + 9x + 14 \checkmark$$

$$0 = (x+7)(x+2) \checkmark$$

$$x = -7 \quad \text{or} \quad x = -2 \checkmark$$

N/A } $\checkmark$

4) $x^2 + 5x - 6 > 0$

(3)

$$(x+6)(x-1) > 0 \checkmark$$



$$x < -6 \checkmark \quad \text{or} \quad x > 1 \checkmark$$

If straight to

$x > -6$ $x > 1$ → no marks



Christo

5) i) Solve for k:

$$k - 1 + \frac{15}{k+2} = 5$$

$$\text{LCD: } k+2$$

$$k \neq -2$$

(5)

$$(k-1)(k+2) + 15 = 5(k+2) \quad \checkmark$$

$$k^2 + k - 2 + 15 = 5k + 10$$

$$k^2 - 4k + 3 = 0 \quad \checkmark$$

$$(k-3)(k-1) \checkmark = 0$$

$$k = 3 \quad \checkmark \quad \text{or}$$

$$k = 1 \quad \checkmark$$

ii) Hence solve for x:

$$2x^2 - x - 1 + \frac{15}{2x^2 - x + 2} = 5$$

(5)

$$\text{Let } k = 2x^2 - x$$

$$\therefore k - 1 + \frac{15}{k+2} = 5 \quad \checkmark$$

$$k^2 + k - 2 + 15 = 5(k+2)$$

$$k^2 + k - 2 + 15 = 5k + 10$$

$$k^2 - 4k + 3$$

$$(k-1)(k-3)$$

$$k=1 \quad k=3$$

$$\therefore k = 3$$

$$2x^2 - x = 3 \quad \checkmark$$

$$2x^2 - x - 3 = 0$$

$$(2x-3)(x+1) = 0$$

$$x = 3/2 \quad \text{or} \quad x = -1$$



$$k = 1$$

$$2x^2 - x = 1 \quad \checkmark$$

$$2x^2 - x - 1 = 0$$

$$(2x+1)(x-1) = 0$$

$$x = -1/2 \quad \text{or} \quad x = 1$$



6) $y = x - 1$ and $y + 7 = x^2 + 2x$

(5)

$$x - 1 + 7 = x^2 + 2x$$

$$0 = x^2 + x - 6$$

$$0 = (x + 3)(x - 2)$$

$$\therefore x = -3 \quad \left\{ \begin{array}{l} y = -4 \end{array} \right. \quad \text{or} \quad x = 2 \quad \left\{ \begin{array}{l} y = 1 \end{array} \right.$$

7) $2^{2x-1} = 4^x - 2$

(5)

$$2^{2x} \cdot 2^{-1} = 2^{2x} - 2$$

$$2^{2x} \cdot 2^{-1} - 2^{2x} = -2$$

$$2^{2x} \left(\frac{1}{2} - 1 \right) = -2$$

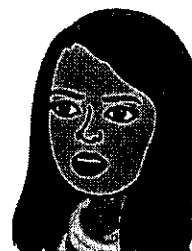
$$2^{2x} \left(-\frac{1}{2} \right) = -2$$

$$2^{2x} = 4$$

$$2x = 2$$

$$x = 1$$

THINKING...



QUESTION 3

13 MARKS

- a) Without solving $3x^2 - x - 4 = 0$, discuss the nature of the roots

(4)

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (-1)^2 - 4(3)(-4) \checkmark \\ &= 1 + \cancel{48} \quad 48 \\ &= \cancel{50} \quad 49 \checkmark\end{aligned}$$

$\therefore$ roots are real, ~~irrational~~, unequal $\checkmark \checkmark$

- b) For which value(s) of r will $2x^2 - 2x - r = 0$ have equal roots?

(3)

$$\begin{aligned}\Delta &= 0 \checkmark \\ b^2 - 4ac &= 0 \\ (-2)^2 - 4(2)(-r) &= 0 \checkmark \\ 4 + 8r &= 0 \\ 8r &= -4 \\ r &= -1/2 \checkmark\end{aligned}$$

Andreg

c) Consider the parabola $y = x^2 + 3$ and the line $y = k$. Find values of k such that the line and parabola will:

1) have two distinct points of intersection.

(4)

$$x^2 + 3 = k \checkmark$$

$$x^2 + 3 - k = 0$$

$$b^2 - 4ac > 0 \checkmark$$

$$0^2 - 4(1)(3 - k) > 0 \checkmark$$

$$-12 + 4k > 0$$

$$4k > 12$$

$$k > 3 \checkmark$$

2) have no points in common.

(2)

$$b^2 - 4ac < 0 \checkmark$$

$$\therefore k < 3 \checkmark$$

HANG IN
THERE



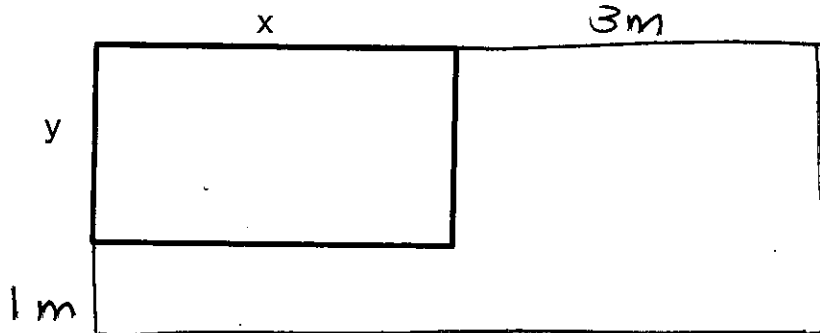
QUESTION 4

7 MARKS



- a) Professor Sprout is wanting to increase the size of her mandrake garden. She wants to increase the length by 3m and the width by 1m. The area of the new garden is now three times larger than that of the original garden.

1) Complete the diagram to illustrate the new and old garden. (1)



2) Determine the original dimensions of the garden, if its area was 6m^2 . (6)

Area of original : $xy = 6$
 $x = 6/y$ ✓

Area of new : $(x+3)(y+1) = 3(6)$ ✓

$$xy + 3y + x + 3 = 18$$

$$\left(\frac{6}{y}\right)y + 3y + \left(\frac{6}{y}\right) - 15 = 0$$
 ✓

$$6 + 3y + \frac{6}{y} - 15 = 0$$

(xy)

$$3y^2 - 9y + 6 = 0$$

$$y^2 - 3y + 2 = 0$$

$$(y-2)(y-1) = 0$$
 ✓

$$\left. \begin{matrix} y=2 \\ x=3 \end{matrix} \right\} \text{✓} \quad \text{or} \quad \left. \begin{matrix} y=1 \\ x=6 \end{matrix} \right\} \text{✓}$$

Dimensions are $3\text{m} \times 2\text{m}$ or $6\text{m} \times 1\text{m}$

QUESTION 5

21 MARKS

a) The n^{th} term of a sequence is given by $T_n = (-2)^{n-1}$

1) Write down the first four terms of the sequence (1)

$$\left. \begin{array}{l} T_1 = 1 \\ T_2 = -2 \\ T_3 = 4 \\ T_4 = -8 \end{array} \right\} \checkmark$$

2) What type of sequence is this? (1)

Geometric $\checkmark$

3) Which term will be equal to 64? (2)

$$\begin{aligned} 64 &= (-2)^{n-1} \\ (-2)^6 &= (-2)^{n-1} \\ 6 &= n-1 \\ n &= 7 \checkmark \end{aligned}$$

4) Calculate the sum of the first 20 terms (3)

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \checkmark \\ S_{20} &= \frac{1 [(-2)^{20} - 1]}{-2 - 1} \\ &= -349525 \checkmark \end{aligned}$$

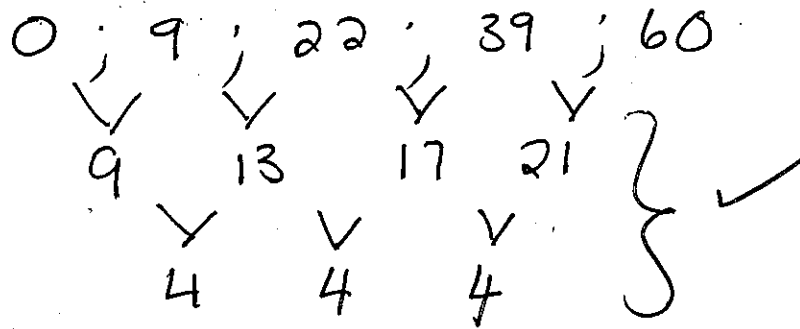


b) Given the number pattern:

0; 9; 22; 39; 60

1) Show that this is a quadratic number pattern.

(1)



2) Find the n^{th} term.

(4)

$$2a = 4$$
$$a = 2 \checkmark$$

$$3a + b = 9$$
$$3(2) + b = 9$$
$$b = 3 \checkmark$$

$$a + b + c = 0$$
$$2 + 3 + c = 0$$
$$c = -5 \checkmark$$

$$T_n = 2n^2 + 3n - 5 \checkmark$$



Marij

c) Write the following in sigma notation:

(4)

$$2; \frac{4}{3}; \frac{8}{9}; \dots; \frac{128}{729}$$

$$T_n = 2 \left(\frac{2}{3} \right)^{n-1} \checkmark$$

$$\frac{128}{728} = 2 \left(\frac{2}{3} \right)^{n-1} \checkmark$$

$$\frac{64}{729} = \left(\frac{2}{3} \right)^{n-1}$$

$$\left(\frac{2}{3} \right)^6 = \left(\frac{2}{3} \right)^{n-1}$$

$$6 = n-1$$

$$7 = n \checkmark$$

$$\therefore \sum_{n=1}^7 2 \left(\frac{2}{3} \right)^{n-1} \checkmark$$

d) Determine the number of terms in the following series:

(5)

$$-1 + 3 + 7 + \dots = 594$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$594 = \frac{n}{2} [2(-1) + (n-1)(4)] \checkmark$$

$$1188 = n(-2 + 4n - 4)$$

$$1188 = 4n^2 - 6n$$

$$0 = 4n^2 - 6n - 1188$$

$$0 = 2n^2 - 3n - 594 \checkmark$$

$$0 = (n-18)(2n+33) \checkmark$$

$$n = 18 \checkmark \quad \text{or} \quad n = -\frac{33}{2}$$

N/A.

QUESTION 6

18 MARKS

a) Given $f(x) = \frac{8}{x-8} + 4$ and $g(x) = 2^{x-1} - 2$

1) Write down the equations of the asymptotes of f and g

(3)

$$f: \begin{aligned} x &= 8 \checkmark \\ y &= 4 \checkmark \end{aligned}$$

$$g: y = -2 \checkmark$$

2) For which values of x is $f(x) = 0$

(2)

$$0 = \frac{8}{x-8} + 4 \checkmark$$

$$-4 = \frac{8}{x-8}$$

$$-4(x-8) = 8$$

$$-4x + 32 = 8$$

$$-4x = -24 \checkmark$$

$$x = 6 \checkmark$$

3) Determine the value of p , if $A(0; p)$ lies on the graph of f .

(2)

$$p = \frac{8}{0-8} + 4 \checkmark$$

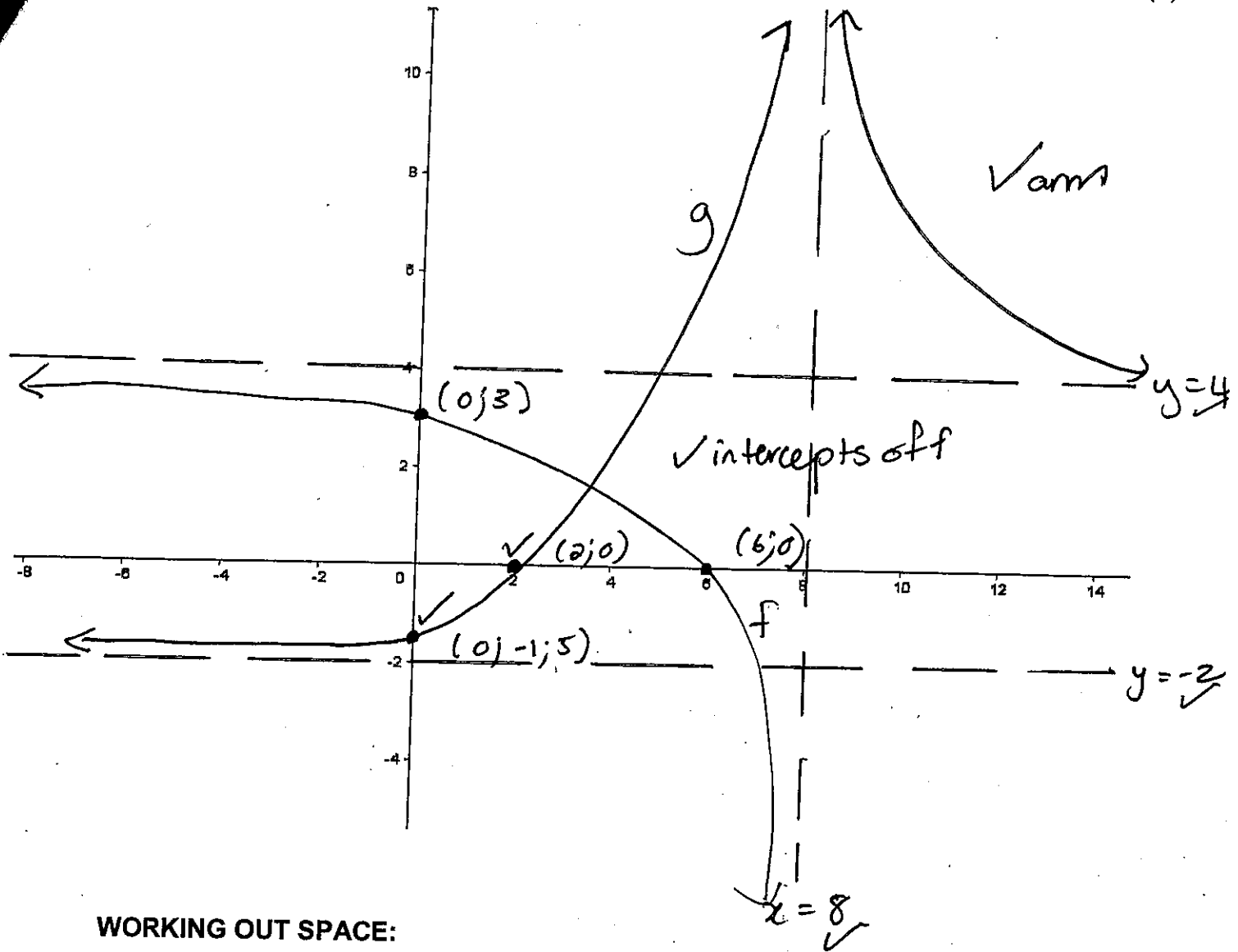
$$p = -1 + 4$$

$$p = 3 \checkmark$$

Grant

4) Sketch the graph of f and g on the grid provided

(7)



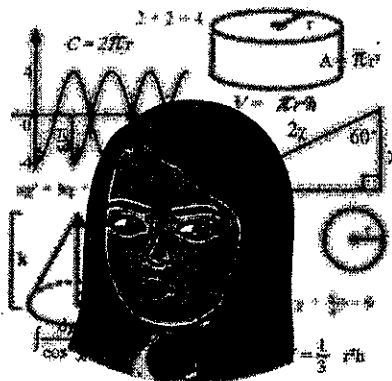
WORKING OUT SPACE:

- 5) Write down the equation of the axis of symmetry of f that has an increasing gradient (2)

$$\begin{aligned}
 y &= x + c \\
 4 &= 8 + c \\
 -4 &= c \\
 \therefore y &= x - 4
 \end{aligned}$$

- 6) Write down the domain and range of g (2)

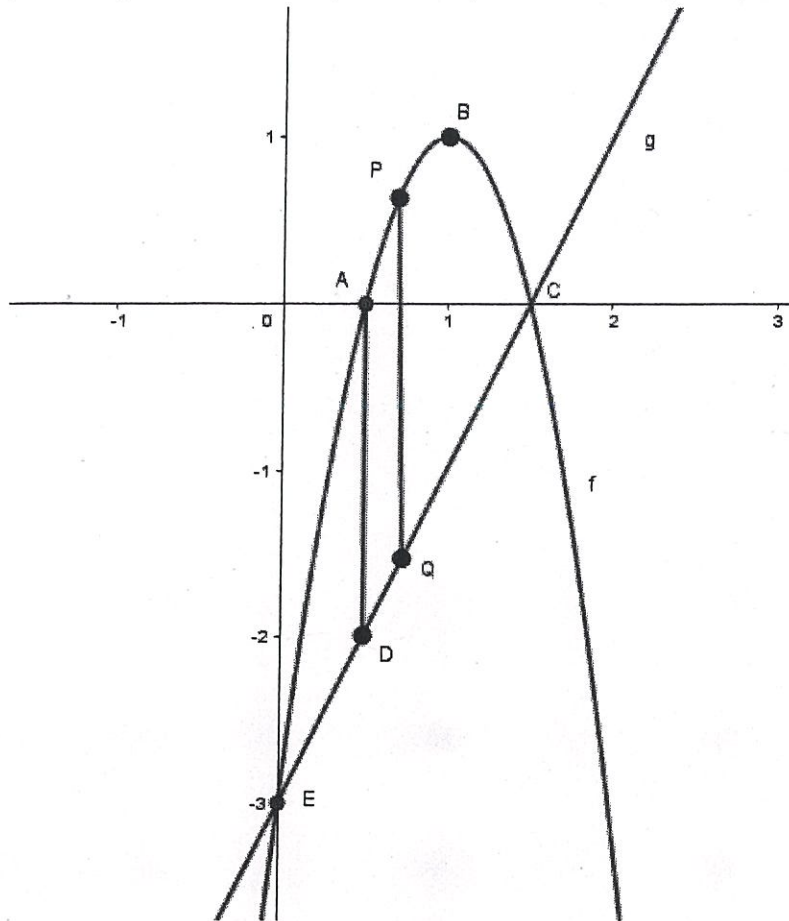
$$\begin{aligned}
 \text{Domain: } & x \in \mathbb{R} \\
 \text{Range: } & y > -2
 \end{aligned}$$



QUESTION 7

19 MARKS

- a) The graph of $f(x) = a(x+p)^2 + q$ and $g(x) = 2x - 3$ are shown in the figure. $B(1; 1)$ is the turning point of f . A and P are points on f and D and Q are points on g .



- 1) Find the equation of f .

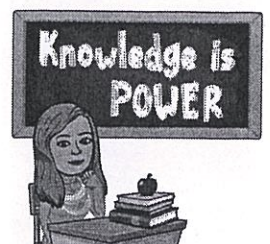
(3)

$$f(x) = a(x-1)^2 + 1 \checkmark$$

$$-3 = a(0-1)^2 + 1 \checkmark$$

$$-4 = a \checkmark$$

$$\therefore f(x) = -4(x-1)^2 + 1 \checkmark$$



- 2) Determine the co-ordinates of A and C

(4)

$$0 = -4(x-1)^2 + 1 \checkmark$$

$$4(x-1)^2 = 1$$

$$(x-1)^2 = \frac{1}{4}$$

$$x-1 = \pm \frac{1}{2} \checkmark$$

$$x = \frac{1}{2} \text{ or } x = \frac{3}{2}$$

$$A \left(\frac{1}{2}; 0 \right) \checkmark$$

$$C \left(\frac{3}{2}; 0 \right) \checkmark$$

- 3) For which values of x is:

i) $f(x) \geq 0$

(2)

$$\frac{1}{2} \leq x \leq \frac{3}{2} \checkmark$$

ii) $f(x) < g(x)$

(2)

$$x < 0 \text{ or } x > \frac{3}{2} \checkmark$$

- 4) Find the length of AD if $AD \perp$ x-axis.

(2)

$$g\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) - 3$$

$$= -2$$

$$\therefore AD = 2 \text{ units} \checkmark$$

Nathan

5) Find the maximum length of PQ

(6)

$$PQ = -4(x-1)^2 + 1 - (2x-3) \checkmark$$

$$= -4(x^2 - 2x + 1) + 1 - 2x + 3$$

$$= -4x^2 + 8x - 4 + 1 - 2x + 3$$

$$= -4x^2 + 6x \checkmark$$

$$= -4\left(x^2 + \frac{3}{2}x\right)$$

$$= -4\left[\left(x + \frac{3}{4}\right)^2 - \frac{9}{16}\right]$$

$$= -4\left(x + \frac{3}{4}\right)^2 + \frac{9}{4} \checkmark$$

$$\begin{aligned} & -4\left(x^2 - \frac{3}{2}x\right) \\ & -4\left(x^2 - \frac{3}{2}x + \frac{9}{16} - \frac{9}{16}\right) \\ & -4\left[\left(x - \frac{3}{4}\right)^2 - \frac{9}{16}\right] \\ & = -4\left(x - \frac{3}{4}\right)^2 + \frac{9}{4} \end{aligned}$$

$$\therefore \text{Max} = \frac{9}{4} \text{ units}$$

$\frac{9}{4}$ units

6) Write down the equation of $h(x)$ if it is the reflection of f across the y-axis.

(1)

$$y = -4(x+1)^2 + 1 \checkmark$$

$$\begin{aligned} & x^2 + 2x + 1 \\ & (x+1)^2 \end{aligned}$$

